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Canadian Bioinformatics Workshops

Module 7: Conclusions

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DALHOUSIE UNIVERSITY
FACULTY OF MEDICINE

汕头大学医学院
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Learning Objectives

- Understand MLOps (Machine Learning Operations).
- Save and reload trained models.
- Deploy models.
- Challenges in ML Production

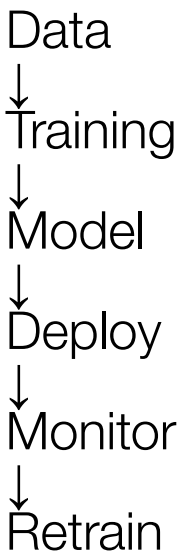
What is MLOps?

Traditional software development cycle:

Write code
↓
Test
↓
Deploy
↓
Maintain

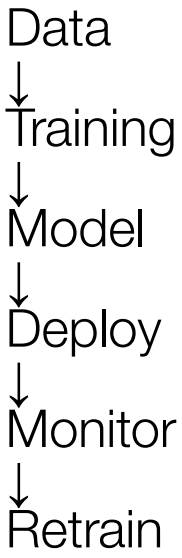
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ML software development cycle:



What is MLOps?

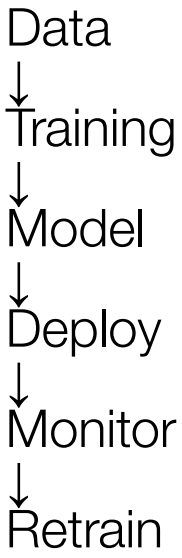
ML software development cycle:



- The code may stay the same;
- **The data changes;**

What is MLOps?

ML software development cycle:



- The code may stay the same;
- **The data changes;**

MLOps is a set of practices to automate, deploy, monitor and maintain ML models through their lifecycle.

What is MLOps?

Detailed ML lifecycle

Collect data	Train model	Deploy
↓	↓	↓
Clean data	Evaluate	Monitor
↓	↓	↓
Feature engineering	Save model	Retrain

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Detailed ML lifecycle

Collect data	Train model	Deploy
↓	↓	↓
Clean data	Evaluate	Monitor
↓	↓	↓
Feature engineering	Save model	Retrain

Where can failures happen?

Model Persistence

- Let's suppose we train a model:
  - model.fit(X_train, y_train)
- What happens when Python closes?

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  Let's jump to the notebook.

Coffee Break & Networking Session

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